Week 2:

Day 4— Task 1

Question:

1. Bank Management System

Objective

Design a Python program for a Bank Management System that demonstrates:

- Abstraction using an abstract class *BankAccount*
- Inheritance using *SavingsAccount*
- Encapsulation by protecting account balance
- Polymorphism by overriding the *calculate_interest()* method
- Use of class and constructor

Sample Input

- Name: Ravi
- Balance: 10000

Sample Output

- Balance: 10000
- Interest: 400.0

Program (BankManagementSystem.py):

```
from abc import ABC, abstractmethod
class BankAccount(ABC):
    def __init__(self, name, balance):
        self.time = 1
        self.interest_rate = 0.04
        self.name = name
        self.__balance = balance
```

Nivetha (CSBS)

```
    @abstractmethod
    def calculate_interest(self):
        pass

class SavingsAccount(BankAccount):
    def __init__(self, name, balance):
        super().__init__(name, balance)
        self.__balance = 0
        self.time = 1
        self.interest_rate = 0.04

    def get_input(self, name, balance):
        self.name = name
        self.__balance = balance

    def calculate_interest(self):
        interest = self.__balance * self.interest_rate * self.time
        return interest

class CheckingAccount(BankAccount):
    def __init__(self, name, balance):
        super().__init__(name, balance)
        self.name = name
        self.__balance  = balance
```

Nivetha (CSBS)

```
    def calculate_interest(self):
        self.interest = self.__balance * self.interest_rate * self.time
        return float(self.interest)

name = input("Enter the name: ")
balance = int(input("Enter the balance: "))
account1 = CheckingAccount(name, balance)
print('Balance: ', account1._BankAccount__balance)
print('Interest: ', account1.calculate_interest())
```

Output:

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS                                   Python

PS C:\Users\Nivetha B\Downloads\nive-python-training> & "C:\Users\Nivetha B\AppData\Local\Python\pythoncore-3.14-64\python.exe" "c:/Users/Nivetha B/Downloads/nive-python-training/BankManagementSystem.py"
Enter the name: ravi
Enter the balance: 10000
Balance:  10000
Interest:  400.0
PS C:\Users\Nivetha B\Downloads\nive-python-training>
```